Acta Tropica 127 (2013) 1–5

Contents lists available at SciVerse ScienceDirect

Acta Tropica

journal homepage: www.elsevier.com/locate/actatropica

Epidemiologic study and molecular detection of *Leishmania* and sand fly species responsible of cutaneous leishmaniasis in Foum Jamâa (Azilal, Atlas of Morocco)

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ARTICLE INFO

Article history:
Received 13 December 2012
Received in revised form 25 February 2013
Accepted 10 March 2013
Available online 21 March 2013

Keywords:
Cutaneous leishmaniasis
ITS1-PCR
Leishmania tropica
Phlebotomus sergenti
Morocco

ABSTRACT

The region of Foum Jamâa (province of Azilal) has become endemic for cutaneous leishmaniasis (CL) since 2006. The objective of this study was to investigate molecular identification of the etiological agent of CL in this region; we also carried out an entomological survey of Phlebotomine sand flies (Diptera: Psychodidae) in this focus to study the sand fly fauna, species composition, and the monthly prevalence of sand flies during 1 year. In the period between 2009 and 2010, skin scrapings spotted on glass slides were collected from 119 patients, aged from 9 months to 70 years old, who came from 43 localities distributed in 3 sectors in Foum Jamâa (FJ). The ITS1 PCR-RFLP was used to identify the *Leishmania* parasite responsible for the recent cases of CL in FJ. Our results revealed that the disease is caused by *L. tropica*. No significant association was observed between gender and the rate of CL in presenting patients, while the highest rate of positive lesions was found in the age group of 9 years old or under (86.67%). In this study, we found also that *L. tropica* infection mostly caused single lesions (67.90%) that were located in the face (96.30%). Morphological identification was performed on a total of 1152 sand flies (23% females and 77% males) collected by sticky paper traps. 57% of the total collected flies were identified as *Phlebotomus (Paraphlebotomus) sergenti* (*Parrot*).

1. Introduction

In Morocco, three species of *Leishmania* are endemic, causing human cutaneous leishmaniasis (CL). *L. major* is responsible of zoonotic CL and is localized in areas south of the Atlas Mountains (Rioux et al., 1986) where regular epidemics, with more than 2000 cases are reported (Ministry of Health, 2008). In the North of the country, some sporadic cases of CL due to *L. infantum* were observed (Lemrani et al., 1999; Rhajaoui, 2009). *L. tropica* has the largest geographic distribution and is considered a major public health threat (Ministry of Health, 2008); this form was reported for the first time in 1989 (Marty et al., 1989), since this first case a large eco-epidemiological study led to the detection of a large focus in central and southern areas of the country. Recently, Morocco is known for the emergence of several new foci in the north Guessous-Idrissi (Guessous-Idrissi et al., 1997; Rhajaoui et al., 2004, 2007) and *L. tropica* has been found in some regions previously known for transmission of *L. major* (Rhajaoui, 2009). *L. tropica* is considered to be purely anthroponotic, however, more recently it has become clear that in some cases *L. tropica* could be zoonotic (Guessous-Idrissi et al., 1997).

In the province of Azilal, FJ region has become an epidemic focal point for CL.

It seems obvious that FJ was free from CL before 1990; the first cases recorded by the provincial delegation of health go back to 1986 and 1987 in Tanant, at 16 km from FJ. Early in 2000, several provincial centers declared some scattered cases in FJ. During the period 2006–2009, we registered about 500 cases of CL in this region (Arroub et al., 2012), the efforts done by the Ministry of Health between 2006 and 2009 have stabilized the number of cases in the region, nevertheless, the disease is still persistent and the number of cases can increase at any time.

In our previous work focused on the eco-epidemiological and socioeconomic study, we have described CL in FJ as a rural domestic form, characterized by affecting the whole family nucleus, due to the fact that dwellings are located near the natural focus of transmission; thus propitiating the vector's arrival in the house (Arroub et al., 2012). Based on clinical symptoms, we suggested that the

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0001-706X/$ – see front matter
http://dx.doi.org/10.1016/j.actatropica.2013.03.005

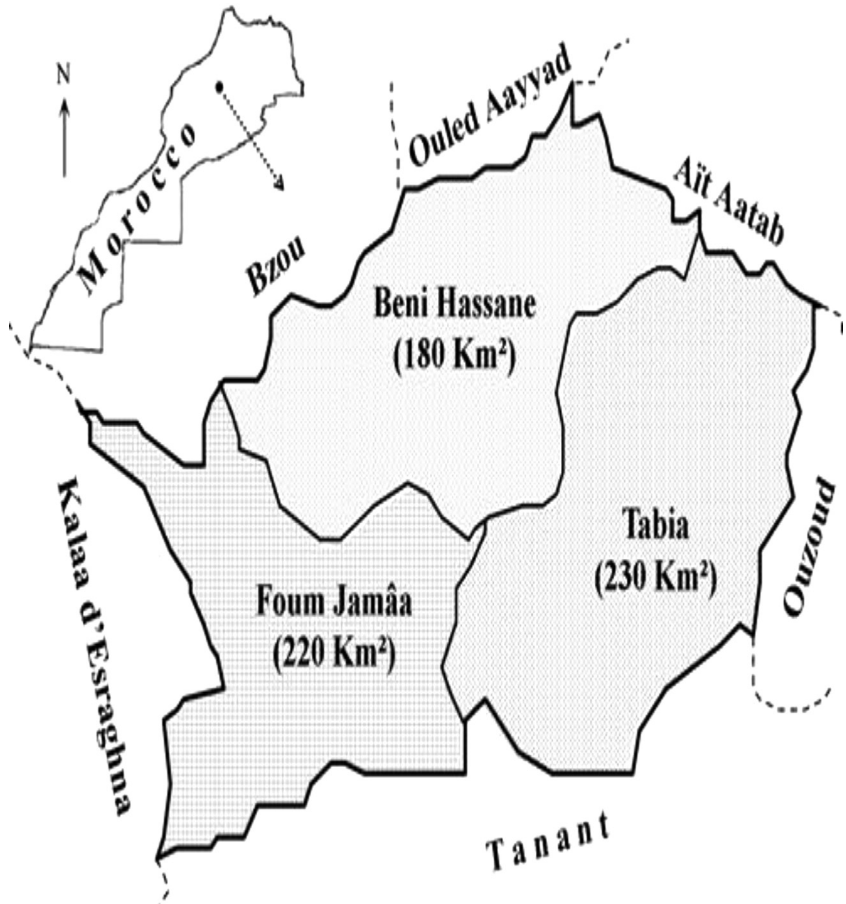

Fig. 1. The location of sampling zones, the surfaces of the three sectors (Beni Hssan, Foum Jamâa, Tabia) are indicated in between bracket below each corresponding sectors name.

disease may be caused by *L. tropica*; however, accurate identification of parasite and sand fly species is crucial to enable adequate treatment and appropriate public health control measures. Accepted standard diagnosis involves isolation of parasites either microscopically, or by culture, but *Leishmania* species are similar and their morphological species identification is not possible (Shahbazi et al., 2008). Currently, multilocus enzyme electrophoresis (MLEE) is the gold standard technique for the identification and classification of isolates of *Leishmania*; however, this method is quite slow, laborious, and costly because it requires culturing and obtaining the profile of 10–20 different enzymes. Lately, several PCR-based methods have proven to be highly sensitive and specific, as compared to standard methods, and are considered valuable for diagnosis (Akhavan et al., 2010; Azizi et al., 2006; Parvizi and Ready, 2008). These molecular methods are particularly useful for identification of *Leishmania* species directly on clinical samples without the need for prior cultivation (Schonian et al., 2003; Vega-Lopez, 2003). The PCR targets include the kDNA minicircles, the small subunit rRNA, the spliced leader sequence, or the miniexon gene. Typing methods, such as restriction analysis of the internal transcribed spacer (ITS) region (Cupolillo et al., 1995), PCR using non-specific primers (Noyes et al., 1996), and direct DNA sequencing and single-strand conformation polymorphism analysis, have been employed.

ITS1 sequences have been extensively studied in *Leishmania* because of their high sensitivity to detect *Leishmania* in clinical samples and their adequately polymorphic to differentiate strains at least to the species level (Schonian et al., 2001). In the present study, we selected the ITS1-PCR method for detecting *Leishmania* in skin scrapings spotted on glass slides; the ITS1 PCR-RFLP was used to identify the *Leishmania* parasite responsible for the recent cases of CL in FJ. Considering the importance of the vector in the transmission of the CL cycle, an entomological survey was carried out in this focus to study the sand fly fauna, species composition, and the monthly prevalence of sand flies during 1 year.

2. Materials and methods

2.1. Study sites

Studies were conducted in 3 adjacent sectors in FJ region, the province of Azilal, High Atlas of Morocco (Fig. 1). The three sectors are Beni Hassan, Foum Jamâa and Tabia, they comprise 43 localities with a total area of about 63,000 ha. It has a population of 30,000 persons. This region is located near the Western High Atlas National Park (32°08′ N, −6°60′ W) at different altitude from 523 to 1086 m. The maximum and minimum mean monthly temperatures were respectively 45 °C (August) and 10 °C (December), and the rainfall is relatively low with two peaks; one in autumn (November to December) and the second in spring (March and April). The vegetation is rare and mainly dominated by cactus, jujube plant, lentils, olive, and almond trees.

2.2. Collection and identification of sand flies

The collections were carried out during 1 year on 17 stations, representing 8 main different biotopes in the region of FJ (henhouse, stable, stable henhouse, cave, ruined house, dwelling, bridge, windows, outdoor) and regularly distributed to cover all supervised area. Sand flies were collected bi-weekly using sticky paper (21 × 27.3) impregnated with grapes oil. The traps installed after sunset were collected before sunrise in the catching sites. Collected sand flies were placed in glass vials containing 70% ethanol. After sex determination, all collected sand flies were mounted on glass slides, using Canada balsam, and were identified by morphological taxonomic keys according to Boussaa (2008).

2.3. Sampling and diagnosis

Tissue samples were taken from 119 patients with suspected CL, who had consulted the health centers in Foum Jamaa during 2009–2010. Before sampling and for each patient we filled a questionnaire with information about the patient (name, age, gender, address) and the lesion (number, localization and date of apparition). From the edge of lesions, smears were prepared, fixed with absolute methanol, and then stained with Giemsa. The whole slide was analyzed with a 100× immersion objective. All of the slides were examined twice before confirming or determining a negative result. A total of 81 slides were subjected to DNA extraction and molecular identification of the causative *Leishmania* species.

Informed written consent was obtained from patients or from parents if patient is minor. Approval for the study was provided by the Ethical Committee of Institut Pasteur of Morocco.

2.4. DNA extraction and PCR analysis

DNA was extracted from stained slides. To each slide, we added 250 μl lysis buffer (50 mM NaCl, 50 mM Tris, 10 mM EDTA, pH 7.4, 1% v/v, Triton X-100 and 100 mg of proteinase K per ml). Cell lysis was accomplished after incubation overnight at 60 °C. Lysates were then subjected to phenol–chloroform extraction, as described by Van Eys et al. (1992). DNA was purified by Qiagen kit according to the instructions of the manufacturer. A final volume of 30 μl obtained were kept at −20 °C until used.

A PCR-restriction fragment length polymorphism (RFLP) approach was applied for the detection and identification of the *Leishmania* parasites. The ribosomal internal transcribed spacer 1 (ITS1) was amplified using the primer pair L5.8S and LITSR (Schonian et al., 2003). Amplicons were analyzed on 1.5% agarose gels by electrophoresis and visualized by UV light. A reaction was considered positive when a band of the correct sizes (300- to 350-bp) was observed. A negative and positive control containing distilled water and DNA of *L. infantum*, respectively, were included during PCR to ensure reliability, validity and to check for possible contaminations of the amplification reactions. PCR product was digested with the restriction endonuclease *HaeIII* (Invitrogen). The resultant fragments were separated by electrophoresis on 3% agarose gels and compared with those of WHO reference strains of *L. major* (MHOM/SU/73/5ASKH), *L. tropica* (MHOM/SU/74/K27) and *L. infantum* (MHOM/TN/1980/IPT1).

3. Results

3.1. Entomological survey

During January–December 2010, 886 males (76.91%) and 226 female (23.09%) sand flies were collected in 1152 phlebotomine specimens. The sex-ratio indicated that more males were collected than females; the male/female ratio was 3.3:1. Overall, six species were identified, consisting of five *Phlebotomus* spp. and one *Sergentomyia* (Table 1). The most dominant species was *P. (Paraphlebotomus) sergenti* (Parrot) (57%), followed by *P. (Larroussius) longicuspis (Nitzulescu)* (23%). The changes in seasonal prevalence of *P. sergenti* are shown in Fig. 2. The population build-up of *P. sergenti* was started in March–April and reached a low level in December (due to the severe weather conditions) with two peaks in June and September. During 2010, the climate conditions in this area were very hot in the summer. The mean maximum and minimum monthly temperatures were 45 °C in August and 10 °C in December, respectively, and the rainfall was relatively low with two peaks; one in autumn (60 mm) (October–December) and the second in spring (100 mm) (March and April).

Table 1
Sand fly species collected in Foum Jamâa region (Azilal, Atlas of Morocco) and their relative abundance.

Sandfly species	Male	Female	Total	%
P. sergenti	486	176	662	57
P. longicuspis	229	31	260	23
P. papatasi	51	24	75	7
P. perniciosus	48	01	49	4
P. chabaudi	01	00	01	0
S. minuta	71	34	105	9
Total	886	266	1152	100

3.2. Active cases survey

From the three major sectors of FJ area Beni Hassan (BH), Foum Jamâa (FJ) and Tabia (T), a total of 119 patients of both sexes and different age groups were included in the study.

Data was analyzed with the non parametric χ^2 test. The repartition of positives cases according to both the sector (BH: 35.24%; FJ: 31.43%; T: 33.33%; $p = 0.892$) and the sex (Male 42.86%; Female 57.14%; $p = 0.143$) was not significant. The age distribution at illness onset ranged from 9 months to 70 years (median age 6.61 years), the disease has been found very significantly prevalent in the age group of 9 years (86.67%; $p < 0.05$) or under (Table 2). Cutaneous lesions were mostly single (67.90%; $p < 0.05$) and located on the face (95.89%).

Overall, CL diagnosis was confirmed in 105 out of 119 suspected cases by microscopic examination and/or ITS1-PCR. *Leishmania* amastigotes were identified in 98 out of 119 skin smears (82.35%). *Leishmania* DNA was detected in 78 of 81 samples examined by ITS1-PCR (96.20%). Only 71 out of 81 samples amplified by PCR were also positive by microscopic examination, seven patients who were negative by microscopic analysis were proven to be positive by the molecular assay.

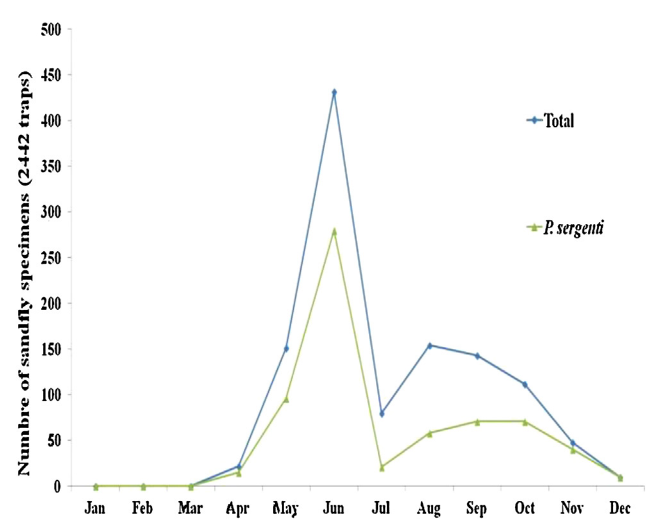

Fig. 2. Seasonal prevalence of sand flies species in Foum Jamâa region (province of Azilal, Atlas of Morocco).

Table 2
The number of positive cutaneous leishmaniasis cases (positive smears and/or + ITS1-PCR) by age in Foum Jamâa region (Azilal, Morocco) recorded from 2009 to 2010.

Age group (years)	Number of positive cases
0–9	**91**
10–19	06
20–29	02
30–39	00
40–49	02
50 & above	04
Total	105

The significance of bold values is the age group 0 - 9 years represents the most recorded number of cases of CL in this study.

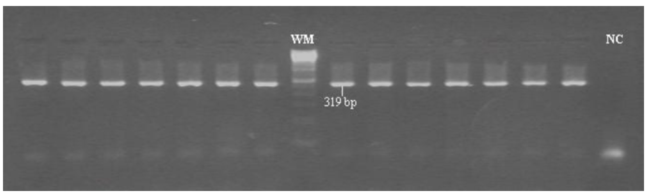

Fig. 3. The results of PCR-based amplication of DNA extracted from Giemsa-stained lesion smears from Foum jamaà in 2009–2010. The bands shown on 1.5% agarose gel stained with ethidium bromide. WM = size marker, NC = negative control. All other lanes correspond to clinical materials from patients.

The *Leishmania* parasites in 78 clinical samples amplified by PCR were identified as *L. tropica* by their typical restriction profiles (Fig. 3). ITS1-PCR produced one single band, which was 300-bp in size. The RFLP analysis revealed two fragments (185- and 57-bp) (Fig. 4), specific for *L. tropica*.

4. Discussion

This is the first report on molecular identification of *L. tropica* as the causative agent of CL in the FJ region. Frequently, *Leishmania* species are identified based on their geographical distribution and on clinical manifestations of the resulting disease. Nevertheless, in endemic regions which may have co-existence of multiple species of *Leishmania*, geographical origin is not an adequate criterion (Rhajaoui, 2009), the characterization of the infecting species based on clinical symptoms is not crucial, since clinical spectrum of CL is broad, symptoms can vary and may be confused with other etiologic agents. On the other hand, the characterization of *Leishmania* species in clinical infections is important, as different species may require distinct treatment regimens, hence the necessity to identify *Leishmania* parasite with certainty.

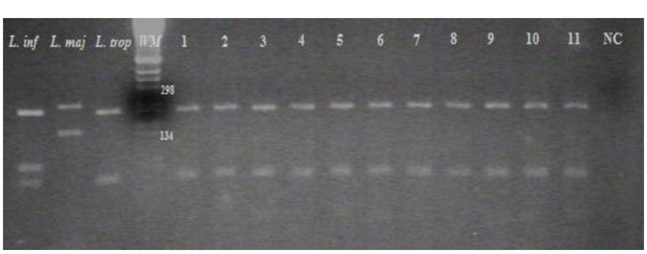

Fig. 4. Molecular identification of causative CL species. Restriction fragment length polymorphism (RFLP) analysis of the amplified internal transcribed spacer 1 region (ITS1) digested with restriction enzyme *HaeIII* and analyzed by electrophoresis on 2.5% agarose gel. Three reference strains were used for comparison: *L. inf* = *L. infantum* (MHOM/TN/1980/IPT1); *L. maj* = *L. major* (MHOM/SU/73/5ASKH); *L. trop* = *L. tropica* (MHOM/SU/74/K27). WM = molecular size marker; NC = negative control. All other lanes show digested PCR product from clinical materials; lanes 1–11 = *L. tropica*.

ITS1-PCR-RFLP used in this study enables easy species identification in all PCR-positive cases. Species identification by this molecular method is clearly less laborious and less costly than that by isoenzyme typing. The ITS1- PCR-RFLP is easily applied in endemic areas, since it is sufficiently sensitive to detect *Leishmania* directly in stained smears and is able to identify all medically relevant species groups (Schonian et al., 2003).

CL in FJ affects both male and female and all age groups, even if the disease was more frequent in young people (9 years or less). The lesions caused by *L. tropica* were dry, small and mainly single (67.90%), while in Taza a northern Moroccan focus, patients exhibited severe, large and multiple lesions, which could be related to the virulence of the parasite having been enhanced by the lack of immunity of the people in this new focus (Guessous-Idrissi et al., 1997). The difference on the clinical manifestations and the age group affected by *L. tropica*, in the northern focus, and the southern one suggest that in FJ the parasite had been present before the outbreak of CL cases; the circulation of the parasite had not been so important to trigger an epidemic, but people living for long time in the area could develop resistance due to their exposure to the parasite.

The appearance of CL due to *L. tropica* has become increasingly important due to many factors. The ecological and demographic changes are probably the most important ones (Ashford, 1999). The increasing urbanization processes in the settlements which have poor housing conditions and sanitation and which are usually over-crowded may facilitate the expansion of the disease. The movement of human and dog population may also act as an important risk factor in the spread of the disease; this might be the cause of the appearance of CL in FJ where the first outbreak appeared in 2006. Previous studies on leishmaniasis in this area have described settlements close to many suitable sources of food for potential animal reservoirs such as dogs. The percentage of positive cases was significantly higher [74.1%, $p \leq 10–4$], where 2 or more dogs were present (Arroub et al., 2012), so that a zoonotic situation should not be excluded, perhaps co-existing with the anthroponotic transmission usually associated with CL due to *L. tropica*. In 1991, Dereure (Dereure et al., 1991) reported canine cutaneous infection due to *L. tropica* in Tanant, a rural area 16 km far from FJ.

We have also demonstrated that *P. sergenti* is the most abundant sand fly species in FJ (57%). *P. sergenti* is the putative vector of *L. tropica* throughout the Middle East and is a proven vector in Morocco (Guilvard et al., 1991), as well as in Saudi Arabia (al-Zahrani et al., 1988) and Afghanistan (Killick-Kendrick et al., 1995).

Researches should be continued to find out if the disease in this region is due to a single or a variety of *L. tropica* genotypes. The disease control requires local stakeholders and health authorities interference.

Conflicts of interest

None.

Funding

This work was supported by EMRO - COMSTECH grant; RAB & GH 10-11/09.

Acknowledgements

The authors thank the Moroccan Ministry of Health (MMH) especially Doctors and nurses of provincial centres of Azilal, who helped in patients sampling. The authors are also very grateful to Dr Lynn Soong for correction of article.

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